

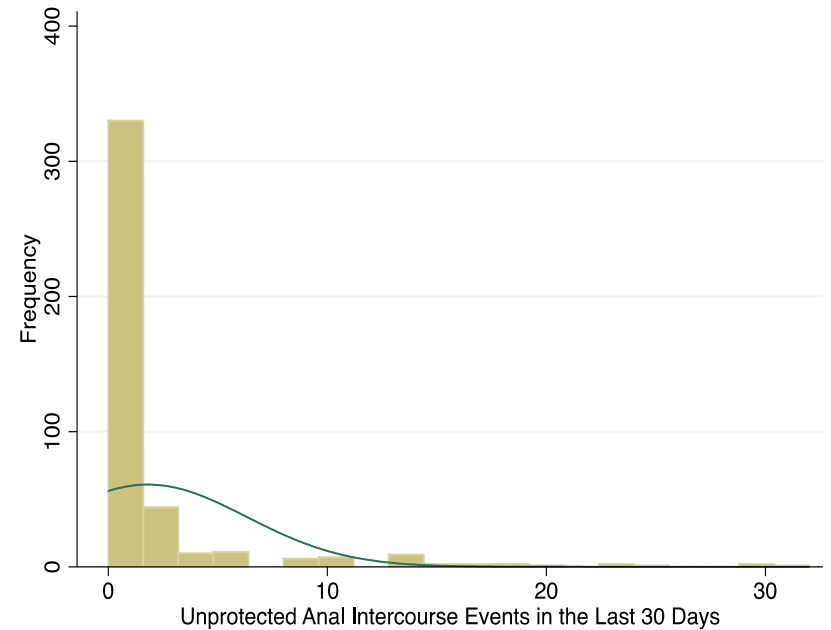
Lectures 11 : Poisson Regression and Other Models

April 17

Learning Objectives

- Poisson Regression models
 - Standard
 - Mixed-Effects
- Other models
 - Binomial Model
 - Negative Binomial (NB) Model
 - Zero-inflated Poisson (ZIP)
- Evaluations

- **Poisson Models:**
Generalized Linear Model (GLM) when the outcome variable is **count data**!



Recall Linear and logistic regression:

- **Data**: $(Y_i, X_{1i}, \dots, X_{pi})$ for $i = 1, \dots, m$
- **Models**: depending on the values of Y 's:
 - **Linear regression** when Y_i are continuous:
 - Assume $Y_i \sim N(\mu_i, \sigma^2)$ [or $Y_i = \mu_i + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$]
 - $\mu_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi}$
 - **Logistic regression** when Y_i are binary or binomial:
 - Assume $Y_i \sim \text{Bernoulli}(n_i, \pi_i)$ and
 - $\log \frac{\pi_i}{1-\pi_i} = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi}$

Generalized Linear Model (GLM)

- **Data:** $(Y_i, X_{1i}, \dots, X_{pi})$ for $i = 1, \dots, m$
- **Generalized Linear Model:** Two steps
 - Specify a distribution for response Y_i (say, with mean μ_i)
 - Specify a link function between μ_i and predictor variables:
$$g(\mu_i) = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi}$$
- **Three Special Cases:** depending on the values of Y 's:
 - When Y_i are continuous: **Linear regression:** assume $Y_i \sim N(\mu_i, \sigma^2)$ [or $Y_i = \mu_i + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$], and $g(\mu_i) = \mu_i$
 - When Y_i are binary or binomial: **Logistic regression.** assume $Y_i \sim \text{Bernoulli}(n_i, \pi_i)$ and $g(\pi_i) = \log \frac{\pi_i}{1-\pi_i}$
 - When Y_i are integer-valued count data: **Poisson regression.** Assume $Y_i \sim \text{Poisson}(\mu_i)$ and $g(\mu_i) = \log \mu_i$

Overview of Poisson Regression Analysis

- y_{it} has a Poisson Distribution:

$$p(y_{it} = r) = \frac{\lambda_{it}^r e^{-\lambda_{it}}}{r!} \quad r = 0, 1, 2, \dots$$

Where λ_{it} is the expected value (mean) of y_{it} .

λ_{it}^r specifies that the mean can vary across person and time.

The Poisson distribution also **assumes** relating to our independent variable....

$$\text{Log}(\lambda_{it}) = \mu_t + \beta x_{it} + \gamma z_i$$

Where x_{it} are the time-varying predictor variables and z_i are the time-invariant predictors. **This loglinear specification ensures that λ will be greater than 0 (non-negative).**

Overview of Poisson Distribution

Suppose Y has a **Poisson(λ)** Distribution with mean λ

- Law of Small Number (Poisson (1781-1840)):

If $Y \sim \text{Binomial}\left(n, p = \frac{\lambda}{n}\right)$ for large n , then $Y \approx \text{Poisson}(\lambda)$.

- $Y \sim \text{Poisson}(\lambda)$ has a probability mass function:

$$P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad \text{for } k = 0, 1, 2, \dots$$

- The mean $E(Y) = \lambda$ and variance $\text{Var}(Y) = \lambda$.

- If $Y_i \sim \text{Poisson}(\lambda_i)$ are independent, then

$$Y_1 + \dots + Y_m \sim \text{Poisson}(\lambda_1 + \dots + \lambda_m)$$

Poisson vs Normal/Gaussian

Suppose Y has a **Poisson(λ)** Distribution with mean λ

Connection of Poisson and Normal/Gaussian distribution:

- **Central Limit Theorem (CLT):** for large mean λ

$$Z = \frac{Y - \lambda}{\sqrt{\lambda}} \approx N(0,1)$$

- **Square Root Transformation:** for large mean λ ,

$$Z = 2\sqrt{Y} \approx N(2\sqrt{\lambda}, 1)$$

- Box-Cox transformation
- **Anscombe Transformation:** $2\sqrt{Y + 3/8}$
- variable-stabilizing transformation

Poisson as Exponential Family

- In Generalized Linear Model (GLM), we consider one-parameter exponential family:

$$f_{\theta}(y) = f_0(y) \exp(\theta y - b(\theta))$$

where $E_{\theta}(Y) = b'(\theta)$ and $Var_{\theta}(Y) = b''(\theta)$

- **For Poisson distribution:** $P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!}$
$$f_{\theta}(y) = \frac{\lambda^y e^{-\lambda}}{y!} = \frac{1}{y!} \exp((\log \lambda)y - \lambda)$$

Thus $\theta = \log \lambda$ and $b(\theta) = \lambda = e^{\theta}$

- Mean $b'(\theta) = e^{\theta} = \lambda$
- Variance $b''(\theta) = e^{\theta} = \lambda$
- GLM for Poisson regression

R code for Poisson regression

- **Poisson regression in GLM for independent data**

```
pois.glm_model <- glm(Y ~ X1 + ...+ Xp, family = poisson,  
                      data = your_data)  
summary(pois.glm_model)
```

- **Poisson mixed-effects model in “lme4” for longitudinal data**

```
pois.lme_model <- glmer(Y ~ X1+...+Xp + (1 | ID),  
                       data = your_data, family = poisson(link = "log"),  
                       offset = log(size) )  
summary(pois.lme_model)
```

GLM: Poisson Regression for Count Data

- Observe $(Y_1, x_{11}, x_{12}, \dots, x_{1,p-1}),$

$$\begin{matrix} \dots & \dots \\ (Y_n, x_{n1}, x_{n2}, \dots, x_{n,p-1}) \end{matrix}$$

with count responses $Y_i \in \{0, 1, 2, 3, \dots\}$

- Model: Two steps

➤ **Distribution: Y_i are Poisson distributed with mean μ_i ,**

$$\text{i.e., } f(y_i) = P(Y_i = y_i) = \frac{\mu_i^{y_i} e^{-\mu_i}}{(y_i)!}$$

➤ link function:

$$\log(\mu_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_{p-1} x_{i,p-1} = x_i^T \beta$$

[Thus $\mu_i = \exp(x_i^T \beta)$].

The MLE estimator of the $\boldsymbol{\beta}_i$'s

- In this case, y_i follows a Poisson distribution, and thus the log-likelihood function is

$$\begin{aligned}\log L(\boldsymbol{\beta}) &= \sum_{i=1}^n (y_i \log(\mu_i) - \mu_i - \log(y_i!)) \\ &= \sum_{i=1}^n (y_i \mathbf{x}_i^T \boldsymbol{\beta} - \exp(\mathbf{x}_i^T \boldsymbol{\beta}) - \log(y_i!))\end{aligned}$$

- The maximum likelihood estimator (MLE) of $\boldsymbol{\beta}_j$'s

➤ Differentiating with respect to $\boldsymbol{\beta}_j$ gives

$$\sum_{i=1}^n (y_i - \exp(\mathbf{x}_i^T \hat{\boldsymbol{\beta}})) \mathbf{x}_{ij} = 0, \text{ for all } j = 0, 1, \dots, p-1$$

or equivalently, $\mathbf{X}^T \mathbf{Y} = \mathbf{X}^T \hat{\boldsymbol{\mu}}$

[Recall that this is also true for OLS when $\hat{\boldsymbol{\mu}} = \mathbf{X} \hat{\boldsymbol{\beta}}$]

Example: 2x2 Table for GLM Regression

	Vaccine No	Vaccine Yes	Total
Shingles No	14	320	334
Shingles Yes	36	80	116
Total	50	400	450

- **Sampling Scheme #a:** Suppose we observe the hospital for a certain period of time and observed 450 patients.
- Assume we classify patients according to whether they have vaccine and disease (shingles).
- **Question:** Does Vaccine affect the Disease/Outcome?
- How to analyze data?
Estimate the expected # in each category.

Data Format

	Vaccine No	Vaccine Yes	Total
Shingles No	14	320	334
Shingles Yes	36	80	116
Total	50	400	450

- The key is to introduce indicator variables:

$$X_1 = \begin{cases} 0 & \text{if vaccine no} \\ 1 & \text{if vaccine yes} \end{cases}$$

$$X_2 = \begin{cases} 0 & \text{if Shingles no} \\ 1 & \text{if Shingles yes} \end{cases}$$

- Observe (Y_i, X_{i1}, X_{i2}) with Y_i = count for each category, for $i = 1, 2, 3, 4$.
- Here we have $n = 4$ observations!

Data for Linear Regression

- Observe (Y_i, X_{i1}, X_{i2}) with Y = count, X_1 =vaccine (“1” for yes), X_2 =shingles (“1” for good), and we have 4 observations

$(14, 0, 0)$

$(320, 1, 0)$

$(36, 0, 1)$

$(80, 1, 1)$

- **Linear Regression Model:**

- Y_i are normally distributed with mean μ_i

- $\mu_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}$

- Or $Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i$

Linear Regression

- Fitting data $(14, 0, 0)$, $(320, 1, 0)$, $(36, 0, 1)$, $(80, 1, 1)$ to Linear Regression Model:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i$$

- Data

$$Y_1 = 14 = \beta_0 + \quad + \quad + \epsilon_1$$

$$Y_2 = 320 = \beta_0 + \beta_1 + \quad + \epsilon_2$$

$$Y_3 = 36 = \beta_0 + \quad + \beta_2 + \epsilon_3$$

$$Y_4 = 80 = \beta_0 + \beta_1 + \beta_2 + \epsilon_4$$

- So we want to minimize

$$RSS = [14 - \beta_0]^2 + [320 - (\beta_0 + \beta_1)]^2 \\ + [36 - (\beta_0 + \beta_2)]^2 + [80 - (\beta_0 + \beta_1 + \beta_2)]^2$$

- Taking derivatives and setting to 0, we have

$$\hat{\beta}_0 = 79.5, \quad \hat{\beta}_1 = 175.0, \quad \hat{\beta}_2 = -109.0$$

Fitted Values from linear regression

- $\hat{\beta}_0 = 79.5$, $\hat{\beta}_1 = 175.0$, $\hat{\beta}_2 = -109.0$
- $Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i$

	Vaccine No	Vaccine Yes	Total
Shingles No	14 (79.5)	320 (254.5)	334
Shingles Yes	36 (-29.5)	80 (145.5)	116
Total	50	400	450

- Does the linear regression model look reasonable to you?

R code for Linear Regression

```
Y <- c(14, 320, 36, 80); X1 <-c(0,1,0,1); X2 <- c(0,0,1,1);  
summary(lm( Y ~ X1+X2))
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	79.5	113.5	0.701	0.611
X1	175.0	131.0	1.336	0.409
X2	-109.0	131.0	-0.832	0.558

Residual standard error: 131 on 1 degrees of freedom

Multiple R-squared: 0.7124, Adjusted R-squared: 0.1372

F-statistic: 1.238 on 2 and 1 DF, p-value: 0.5363

Count Data for Poisson Regression

- Observe (Y_i, X_{i1}, X_{i2}) with Y = count, X_1 =vaccine (“1” for yes), X_2 =shingles (“1” for good), and we have 4 observations

$(14, 0, 0)$

$(320, 1, 0)$

$(36, 0, 1)$

$(80, 1, 1)$

- **Poisson Regression Model:**

➤ Y_i are Poisson distributed with mean μ_i

➤ $\log(\mu_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}$

Maximum Likelihood Estimator (MLE)

- The log-likelihood is

$$\begin{aligned}\log L(\boldsymbol{\beta}) &= \sum_{i=1}^n (y_i \log(\mu_i) - \mu_i - \log(y_i!)) \\ &= 14\beta_0 - e^{\beta_0} - \log(14!) \\ &\quad + 320(\beta_0 + \beta_1) - e^{\beta_0 + \beta_1} - \log(320!) \\ &\quad + 36(\beta_0 + \beta_2) - e^{\beta_0 + \beta_2} - \log(36!) \\ &\quad + 80(\beta_0 + \beta_1 + \beta_2) - e^{\beta_0 + \beta_1 + \beta_2} - \log(80!)\end{aligned}$$

- Next, taking derivatives yields

$$e^{\beta_0}(1 + e^{\beta_1})(1 + e^{\beta_2}) = 14 + 320 + 36 + 80 = 450$$

$$e^{\beta_0 + \beta_1}(1 + e^{\beta_2}) = 320 + 80 = 400$$

$$e^{\beta_0 + \beta_2}(1 + e^{\beta_1}) = 36 + 80 = 116$$

- Hence $e^{\beta_1} = \frac{400}{450 - 400}$, $e^{\beta_2} = \frac{116}{450 - 116}$, $e^{\beta_0} = \frac{50 \times 334}{450}$

and taking the logarithm (with base e)

$$\hat{\beta}_0 = 3.6139, \quad \hat{\beta}_1 = 2.0794, \quad \hat{\beta}_2 = -1.0576$$

Fitted Values for Poisson

- $\log(\mu_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}$ with $\hat{\beta}_0 = 3.6139 = \log\left(\frac{50 \times 334}{450}\right)$, $\hat{\beta}_1 = 2.0794 = \log\left(\frac{400}{50}\right)$, $\hat{\beta}_2 = -1.0576 = \log\left(\frac{116}{334}\right)$

	Vaccine No	Vaccine Yes	Total
Shingles No	14 $(37.1 = \frac{334 \times 50}{450})$	320 $(296.9 = \frac{334 \times 400}{450})$	334
Shingles Yes	36 $(12.9 = \frac{116 \times 50}{450})$	80 $(103.1 = \frac{116 \times 400}{450})$	116
Total	50	400	450

- Poisson regression: **Fitted value = row sum * col sum / total !!!**
- Pearson's Chi-square Test statistics $\sum \frac{(O-E)^2}{E} = 62.8123!$

R code for Poisson Regression

```
Y <- c(14, 320, 36, 80); X1 <- c(0,1,0,1); X2 <- c(0,0,1,1);  
summary(glm( Y ~ X1+X2, family = poisson) )
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.6139	0.1441	25.075	<2e-16 ***
X1	2.0794	0.1500	13.863	<2e-16 ***
X2	-1.0576	0.1078	-9.813	<2e-16 ***

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 474.10 on 3 degrees of freedom

Residual deviance: 54.03 on 1 degrees of freedom

AIC: 83.774

Example: 2x2 Table for GLM Regression (cont.)

	Vaccine No	Vaccine Yes	Total
Shingles No	14	320	334
Shingles Yes	36	80	116
Total	50	400	450

- **Sampling Scheme #b:** Suppose we follow 50 no-vaccine and 400 vaccine 40-year-old patients from the hospital, and then observe whether they have shingles after 10 years.
- Assume we classify patients according to whether they have vaccine and disease (shingles).
- **Question:** Does Vaccine affect the Disease/Outcome?
- How to analyze data?

Data Format for sampling #b

- There are two groups

- Among 50 subjects without vaccine, 36 shingles & 14 no-shingles
- Among 400 subjects with vaccine, 80 shingles & 320 no shingles.

- The key idea:

$$X = \begin{cases} 0 & \text{if no vaccine} \\ 1 & \text{if vaccine} \end{cases} ; \quad Y = \begin{cases} 0 & \text{if no shingles} \\ 1 & \text{if shingles} \end{cases}$$

- Data is (Y_i, X_i) for $i = 1, \dots, 450$ where

- Among 400 observations with $X_i = 1$, we have 80 of $Y_i = 1$ and 320 of $Y_i = 0$;
- Among 50 observations with $X_i = 0$, we have 36 of $Y_i = 1$ and 14 of $Y_i = 0$.

- Here we have $n = 450$ observations!

Data for Simple Logistic Regression

	Vaccine No	Vaccine Yes	Total
Shingles No	14	320	334
Shingles Yes	36	80	116
Total	50	400	450

	$Y_i = 1$	$Y_i = 0$	Sum	
$X_i = 1$	$a = 80$	$b = 320$	400	$\hat{p}_0 = \frac{80}{400} = 0.20$
$X_i = 0$	$c = 36$	$d = 14$	50	$\hat{p}_1 = \frac{36}{50} = 0.72$

Data for Simple Logistic Regression

- Observe (Y_i, X_i) with $X_i = \text{vaccine}$ (“1” for yes), $Y_i = \text{Shingles}$ (“1” for yes), and we have $n = 450$ observations

	$Y_i = 1$	$Y_i = 0$	Sum	
$X_i = 1$	$a = 80$	$b = 320$	400	$\hat{p}_0 = \frac{80}{400} = 0.20$
$X_i = 0$	$c = 36$	$d = 14$	50	$\hat{p}_1 = \frac{36}{50} = 0.72$

- Simple Logistic Regression Model
 - $P(Y_i = 1) = \pi_i, \quad P(Y_i = 0) = 1 - \pi_i$
 - $\log \frac{\pi_i}{1 - \pi_i} = \beta_0 + \beta_1 x_i$

The MLE estimator of the β_i 's

	$Y_i = 1$	$Y_i = 0$	Sum	
$X_i = 1$	$a = 80$	$b = 320$	400	$\hat{p}_0 = \frac{80}{400} = 0.20$
$X_i = 0$	$c = 36$	$d = 14$	50	$\hat{p}_1 = \frac{36}{50} = 0.72$

- The likelihood function is $L(\theta) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i}$

$$= \frac{e^{\beta_0(a+c) + \beta_1 a}}{(1 + e^{\beta_0 + \beta_1})^{a+b} (1 + e^{\beta_0})^{c+d}}$$

- The MLE is $\hat{\beta}_0 = \log\left(\frac{c}{d}\right) = \log\left(\frac{36}{14}\right) = 0.9445$;

$$\hat{\beta}_1 = \log\left(\frac{ad}{bc}\right) = \log\left(\frac{80 \times 14}{320 \times 36}\right) = -2.3308.$$

$$se(\hat{\beta}_1) = \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}} = 0.3389$$

Association ?

	$Y_i = 1$	$Y_i = 0$	Sum	
$X_i = 1$	$a = 80$	$b = 320$	400	$\hat{p}_0 = \frac{80}{400} = 0.20$
$X_i = 0$	$c = 36$	$d = 14$	50	$\hat{p}_1 = \frac{36}{50} = 0.72$

- Is the X associated with Y?
Is β_1 significantly different from 0?

- Test $H_0: \beta_1 = 0$ versus $H_1: \beta_1 \neq 0$ at the $\alpha = 5\%$ level.
- The observed z-statistics is

$$Z_{obs} = \frac{\hat{\beta}_1 - 0}{se(\hat{\beta}_1)} = \frac{-2.3308}{0.3389} = -6.878$$

- Since $|Z_{obs}| \geq z_{\alpha/2} = z_{0.025} = 1.96$, we reject $H_0: \beta_1 = 0$ at the $\alpha = 5\%$ level. **Thus X and Y are significantly associated!!!**

Besides Logistic, Other Tests for 2x2 table

1. Fisher's test for 2x2 table

2. Two-sample Proportion Test:

a) Observe $X = 80 \sim \text{Binomial}(m = 400, p_0)$

$Y = 36 \sim \text{Binomial}(n = 50, p_1)$

b) When testing $H_0: p_0 = p_1$ against $H_1: p_0 \neq p_1$,

Compute $\hat{p}_0 = \frac{X}{m} = 0.20, \hat{p}_1 = \frac{Y}{n} = 0.72, \hat{p} = \frac{X+Y}{m+n} = \frac{116}{450}$

$$T = \frac{\hat{p}_1 - \hat{p}_0}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{m} + \frac{1}{n} \right)}} = 7.925422$$

3. Pearson's Chi-Square Test from Poisson Regression:

$$\sum \frac{(O-E)^2}{E} = 62.81231 \quad (= 7.925422^2 = T^2)$$

Deviance for GLM

- Data: observe $(Y_1, x_{11}, x_{12}, \dots, x_{1,p-1}),$

$$\begin{matrix} \dots & \dots & \dots \\ (Y_n, x_{n1}, x_{n2}, \dots, x_{n,p-1}) \end{matrix}$$

- Generalized Linear Model (M_0): Two steps

- Specify a distribution for Y_i (with mean μ_i and variance $\phi_i v(\mu_i)$)

- Specify a link function

$$g(\mu_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_{p-1} x_{i,p-1}$$

- **Saturated (or full) model:** a model with (large # of parameters) so that the response Y are fitted exactly.

- Given the data, the **deviance** for the model M_0 is defined as

$$Dev(M_0) = -2 \log L(Y; M_0) + 2 \log L(Y; M_{saturated})$$

Deviance for Poisson Regression

- When Y_i are count data: **Poisson Regression**. assume $Y_i \sim \text{Poisson}(\mu_i)$,
 $g(\mu_i) = \log(\mu_i)$

$$-2 \log L(Y; M_0) = -2 \sum_i (-\log(Y_i!) + Y_i \log(\hat{\mu}_i) - \hat{\mu}_i)$$

- Saturated (or full) model**: a model so that the response data are fitted exactly: $\hat{\mu}_i = Y_i$ and

$$-2 \log L(Y; M_{saturated}) = -2 \sum_i (-\log(Y_i!) + Y_i \log(Y_i) - Y_i)$$

- Deviance for Poisson Regression:**

$$Dev(M_0) = -2 \sum_i \left(Y_i \log \left(\frac{\hat{\mu}_i}{Y_i} \right) - \hat{\mu}_i + Y_i \right)$$

- It can be shown that with intercept term, $\sum \hat{\mu}_i = \sum Y_i$, and thus
 $Dev(M_0) = -2 \sum_i Y_i \log \left(\frac{\hat{\mu}_i}{Y_i} \right) = 2 \sum_i O_i \log \left(\frac{O_i}{E_i} \right)$, which is often called the G-statistics (or G-test) with $O_i = Y_i, E_i = \hat{\mu}_i$.

Pearson Chi-Squared Test from Poisson Regression

- With intercept term, $\sum \hat{\mu}_i = \sum Y_i$, and thus

$$Dev(M_0) = -2 \sum_i Y_i \log\left(\frac{\hat{\mu}_i}{Y_i}\right) = 2 \sum_i O_i \log\left(\frac{O_i}{E_i}\right)$$

becomes the G-statistics and $\sum O_i = \sum E_i$.

- Using the Taylor expansion: $(1 + x)\log(1 + x) \approx x + \frac{x^2}{2}$,

$$\begin{aligned} Dev(M_0) &= 2 \sum_i E_i \left(1 + \frac{O_i - E_i}{E_i}\right) \log\left(1 + \frac{O_i - E_i}{E_i}\right) \\ &\approx 2 \sum_i E_i \left[\frac{O_i - E_i}{E_i} + \frac{1}{2} \left(\frac{O_i - E_i}{E_i}\right)^2 \right] = \sum_i \frac{(O_i - E_i)^2}{E_i}, \end{aligned}$$

which is **Pearson's Chi-squared test statistics**.

- In Poisson regression, $e_i^{pear} = \frac{O_i - E_i}{\sqrt{E_i}} = \frac{Y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}$
is often called **Pearson residuals**.

Dispersion Parameter

- Since Poisson distribution has $E(Y) = Var(Y)$ but real data might not satisfy this, we can introduce a dispersion parameter ϕ so that $Var(Y) = \phi E(Y)$.

- In Poisson regression, the dispersion parameter ϕ can be estimated by

$$\hat{\phi} = \frac{\sum_{i=1}^n [(Y_i - \hat{\mu}_i)^2 / \hat{\mu}_i]}{n - p}$$

- The dispersion parameter only affects $var(\hat{\beta})$ and is independent of the point estimation of $\hat{\beta}$ (and $\hat{\mu}_i$).
- In R (and many statistical software), quasi-Poisson regression refers to use $\hat{\phi}$ for CI and testing (F-test).

Longitudinal Count data

- Poisson mixed-effects model
- Other models

Poisson Regression for longitudinal Count Data

- Observe $(Y_{ij}, x_{1ij}, x_{2ij}, \dots, x_{pij})$, for $i = 1, \dots, m, j = 1, \dots, n_i$
with count responses $Y_i \in \{0, 1, 2, 3, \dots\}$
- Poisson (intercept-random-effects) Model: Two steps
 - **Distribution:** Y_{ij} are Poisson distributed with mean μ_{ij} ,
i.e., $f(y_{ij}) = P(Y_{ij} = y_{ij}) = \frac{\mu_{ij}^{y_{ij}} e^{-\mu_{ij}}}{(y_{ij})!}$
 - link function:
 $\log(\mu_{ij}) = \beta_0 + \beta_1 x_{1ij} + \dots + \beta_p x_{pij} + b_i$
where $b_i \sim N(0, \sigma_b^2)$

More Distributions for Count Data

- **Negative Binomial Distribution (NB(p,r)):**

$$P(Y = y) = \binom{y + r - 1}{y} (1 - p)^y p^r$$

➤ It is often used to handle overdispersion in count data

- **Zero-inflated Poisson (ZIP(π, λ)):**

$$P(Y = 0) = \pi + (1 - \pi)e^{-\lambda}$$

$$P(Y = y) = (1 - \pi) \frac{\lambda^y e^{-\lambda}}{y!} \text{ for } y = 1, 2, 3, \dots$$

- R package: “**glmmTMB**” (Generalized Linear Model Mixed Models using Template Model Builder).

One final remark on overdispersion

Besides Negative Binomial Distribution, the over-dispersion can also be modeled or addressed by

- **A mixture of Poisson distributions:**

$$P(Y = y) = \sum_i \pi_i \frac{\lambda_i^y}{y_i!} e^{-\lambda_i} \quad \text{for } y = 0, 1, 2, \dots$$

where $\sum_i \pi_i = 1$.

- **Box-cox transformation:**

$$Y^* = \begin{cases} \frac{Y^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \log(Y) & \text{if } \lambda = 0 \end{cases}$$

The remaining slides are from past years
by Dr. Cook

Key Assumptions of Poisson Regression

- **Assumption 1**

- Regarding DV: The mean = the variance
- Violation of Assumption 1:
 - **Overdispersion**: the variance > mean
 - **Underdispersion**: the mean > variance (we rarely see this)

- **Assumption 2 (independent observations)**

- The events that make up a Poisson distribution are independent (hard not to violate)

NOTES

- Both of these assumptions are often **violated** given certain types of data
 - We have ways to account for overdispersion (alternative models: zero inflated models, overdispersion models)
 - It is often the case that an event occurring may increase (or decrease) the likelihood of the event occurring again (violation of independent observations).

Overdispersion is the most common violation

- An example of overdispersion.
- Overdispersion occurs when the variance is greater than the mean
- We can assess overdispersion by looking at the descriptive statistics

Descriptive Statistics

	n	Mean	Variance	Max	Min
uaiall	430	1.81	20.38	0	32

The variance is greater than the mean, meaning the data are overdispersed.

Example 1: German Health Care Reform and Number of Dr. Visits

- **id:** person id
- **numvisit:** self-reported number of visits to a doctor during the 3 months before the interview
- **reform:** dummy variable for interview being during the year after the reform v. the year before the reform
- **age:** age in years
- **educ:** education in years
- **married:** dummy variable for being married
- **badh:** dummy variable for self reported health (poor vs. well)
- **loginc:** logarithm of household income

Example : Pop Avg. Poisson

```
. poisson numvisit reform age educ married badh loginc summer, irr vce(cluster id)
```

```
Iteration 0:    log pseudolikelihood = -5947.9413
Iteration 1:    log pseudolikelihood = -5942.6967
Iteration 2:    log pseudolikelihood = -5942.6924
Iteration 3:    log pseudolikelihood = -5942.6924
```

Poisson regression	Number of <u>obs</u>	=	2,227
	Wald chi2(7)	=	248.40
	Prob > chi2	=	0.0000
Log pseudolikelihood = -5942.6924	Pseudo R2	=	0.1073

(Std. Err. adjusted for 1,518 clusters in id)

<u>numvisit</u>	IRR	Robust Std. Err.	z	P> z	<u>[95% Conf. Interval]</u>
reform	.8689523	.048284	-2.53	0.011	.7792885 .9689328
age	1.004371	.0033158	1.32	0.186	.9978928 1.01089
educ	.9894036	.0115247	-0.91	0.360	.9670715 1.012251
married	1.042542	.0722475	0.60	0.548	.9101354 1.194212
<u>badh</u>	3.105111	.2642862	13.31	0.000	2.628019 3.668814
<u>loginc</u>	1.160559	.091528	1.89	0.059	.9943449 1.354558
summer	1.010269	.0939635	0.11	0.913	.8419145 1.212288
_cons	.6617582	.3796245	-0.72	0.472	.2149803 2.037042

Example : Pop Avg. Poisson

```
glm numvisit reform age educ married badh loginc summer, family(poisson) link(log) vce(cluster id) eform
```

Generalized linear models		No. of obs	=	2,227
Optimization	: ML	Residual df	=	2,219
		Scale parameter	=	1
Deviance	= 7419.853221	(1/df) Deviance	=	3.343782
Pearson	= 9688.740471	(1/df) Pearson	=	4.366264
Variance function:	$V(u) = u$			[Poisson]
Link function	: $g(u) = \ln(u)$			[Log]
		AIC	=	5.344133
Log pseudolikelihood = -5942.69244		BIC	=	-9685.11

(Std. Err. adjusted for 1,518 clusters in id)

numvisit	IRR	Robust Std. Err.	z	P> z	[95% Conf. Interval]	
reform	.8689523	.048284	-2.53	0.011	.7792885	.9689328
age	1.004371	.0033158	1.32	0.186	.9978928	1.01089
educ	.9894036	.0115247	-0.91	0.360	.9670715	1.012251
married	1.042542	.0722475	0.60	0.548	.9101354	1.194212
badh	3.105111	.2642862	13.31	0.000	2.628019	3.668814
loginc	1.160559	.091528	1.89	0.059	.9943449	1.354558
summer	1.010269	.0939635	0.11	0.913	.8419145	1.212288
_cons	.6617582	.3796245	-0.72	0.472	.2149803	2.037042

Example : Pop Avg. Poisson

Population average interpretation.....

- There was a 13% reduction in number of doctor visits between 1996-1998, controlling for the other covariates in the model. This signals that reforming the health system worked.

. xtpoisson numvisit reform age educ married badh loginc summer, normal irr

Random-effects Poisson regression
Group variable: **id**

Number of obs = **2,227**
Number of groups = **1,518**

Random effects u_i ~ **Gaussian**

Obs per group:
min = **1**
avg = **1.5**
max = **2**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-4643.3608**

Wald chi2(7) = **253.16**
Prob > chi2 = **0.0000**

numvisit	IRR	Std. Err.	z	P> z	[95% Conf. Interval]	
reform	.9547597	.0310874	-1.42	0.155	.895733	1.017676
age	1.006003	.0028278	2.13	0.033	1.000475	1.01156
educ	1.008656	.0127767	0.68	0.496	.9839221	1.034011
married	1.077904	.0595749	1.36	0.175	.9672414	1.201228
badh	2.466573	.1519255	14.66	0.000	2.186076	2.783061
loginc	1.097455	.0747067	1.37	0.172	.9603805	1.254095
summer	.8672783	.0562749	-2.19	0.028	.763707	.9848956
_cons	.5159176	.2645129	-1.29	0.197	.1888714	1.409271
/lnsig2u	-.2016963	.0611823			-.3216115	-.0817812
sigma_u	.9040703	.0276566			.8514575	.9599341

```
. xtmeipoisson numvisit reform age educ married badh loginc summer || id:, irr
```

Refining starting values:

```
Iteration 0:   log likelihood = -4761.4845
Iteration 1:   log likelihood = -4658.2456
Iteration 2:   log likelihood = -4643.3663
```

Performing gradient-based optimization:

```
Iteration 0:   log likelihood = -4643.3663
Iteration 1:   log likelihood = -4643.2823
Iteration 2:   log likelihood = -4643.2823
```

Mixed-effects Poisson regression
Group variable: id

```
Number of obs      =      2,227
Number of groups   =      1518
```

Obs per group:

```
min =      1
avg  =      1.5
max  =      2
```

```
Integration points =      7
Log likelihood = -4643.2823
```

```
Wald chi2(7)      =      253.06
Prob > chi2       =      0.0000
```

numvisit	IRR	Std. Err.	z	P> z	[95% Conf. Interval]	
reform	.9547758	.0310889	-1.42	0.155	.8957463	1.017695
age	1.006003	.0028283	2.13	0.033	1.000475	1.011562
educ	1.008659	.0127792	0.68	0.496	.9839207	1.034019
married	1.077902	.0595839	1.36	0.175	.9672229	1.201245
badh	2.466399	.1519349	14.65	0.000	2.185887	2.782909
loginc	1.09743	.074714	1.37	0.172	.9603425	1.254086
summer	.8672584	.0562778	-2.19	0.028	.7636823	.9848823
_cons	.5159046	.2645398	-1.29	0.197	.1888425	1.409416

Note: _cons estimates baseline incidence rate (conditional on zero random effects).

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]	
id: Identity				
sd(_cons)	.9043832	.0276826	.8517217	.9603006

```
LR test vs. Poisson model: chibar2(01) = 2598.82      Prob >= chibar2 = 0.0000
```

```
. xtmeipoisson numvisit reform age educ married badh loginc summer || id: reform,
covariance(unstructured) irr
```

Refining starting values:

```
Iteration 0:  log likelihood = -4687.5516
Iteration 1:  log likelihood = -4562.6326
Iteration 2:  log likelihood = -4514.5293
```

Performing gradient-based optimization:

```
Iteration 0:  log likelihood = -4514.5293
Iteration 1:  log likelihood = -4513.7312
Iteration 2:  log likelihood = -4513.7299
Iteration 3:  log likelihood = -4513.7299
```

Mixed-effects Poisson regression
Group variable: id

```
Number of obs      =      2,227
Number of groups   =      1518
```

Obs per group:

```
min =      1
avg  =      1.5
max  =      2
```

```
Integration points =      7
Log likelihood = -4513.7299
```

```
Wald chi2(7)      =      241.11
Prob > chi2       =      0.0000
```

numvisit	IRR	Std. Err.	z	P> z	[95% Conf. Interval]	
reform	.9022792	.0483866	-1.92	0.055	.812257	1.002278
age	1.003456	.0028317	1.22	0.222	.997921	1.009021
educ	1.008894	.0128121	0.70	0.486	.9840931	1.034321
married	1.086874	.064118	1.41	0.158	.9681982	1.220097
badh	3.028323	.2322948	14.44	0.000	2.605606	3.519619
loginc	1.135637	.0866473	1.67	0.096	.9778991	1.318817
summer	.9140246	.0741941	-1.11	0.268	.7795846	1.071649
_cons	.4060724	.2305625	-1.59	0.112	.1334451	1.235675

Note: _cons estimates baseline incidence rate (conditional on zero random effects).

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]	
id: Unstructured				
sd(reform)	.9303106	.0561786	.8264688	1.0472
sd(_cons)	.9541108	.0357016	.8866412	1.026715
corr(reform,_cons)	-.4908243	.0506047	-.5835351	-.385484

```
LR test vs. Poisson model: chi2(3) = 2857.93      Prob > chi2 = 0.0000
```

```
. lrtest xtri xtrc
```

Likelihood-ratio test

(Assumption: xtri nested in xtrc)

LR $\chi^2(2)$ = 259.10

Prob > χ^2 = 0.0000

- The random coefficients model is preferred

Summary: Models

- Random effect models
 - Random intercept vs. random coefficient
- GEE (population average model)
 - Tells us what is happening among participants
 - Use with robust standard errors
 - Specify correlation matrix
- Fixed effect models
 - Assessing the causal relationship between a time-vary predictor and outcome
 - Need at least 3 time points
 - Must have significant variation between timepoints
- Hybrid models
 - Influence of time vary and time invariant predictors
- Robust standard errors (produced unbiased estimates)
 - Did not discuss in class much, but important when thinking about population average models
 - Popular in economics and political science

Zero-inflated Poisson

- If there is significant overdispersion in our data, the NB regression procedure is preferred to the Poisson
- However, sometimes there are many, many zero's in your outcome variable
- In these cases the NB model is not sufficient to handle the overdispersion
- Examples
 - # Sexual risk behavior
 - # Violent crimes committed by an individual each year

Zero-inflated Poisson

- Model assumes that there are 2 groups –
 1. A zero group (i.e. structural zero): Will never not be a zero value
 2. A non-zero group: Have a probability of a non-zero count value
- **Group 1:** We use a logit model to determine group membership for those in group 1
- **Group 2:** We use Poisson or Negative Binomial to model the non-zero count for individuals in group 2

Zero-inflated Growth Model

- Extension of of zero-inflated model with the growth factor
- Model estimates growth curve for structural zeros and count simultaneously
- Binary component includes all the observations in the data
- Count component includes all observations but is modeling only those zeros that can be explained by a random Poisson process

Group Based Trajectory Analysis (GBTM) Application:

Understanding Trajectories of Alcohol and Marijuana Use Among Emerging Adult Sexual Minority Men

Cook, S. H., Wood, E. P., Scott, M., Pierce, K. A., Kapadia, F., & Halkitis, P. N. (2020). Predicting trajectories of substance use during emerging adulthood: Exploring the benefits of group-based trajectory modeling for zero-inflated outcomes. *TPM: Testing, Psychometrics, Methodology in Applied Psychology*, 27(3).

Sample Description

- **Data Source: The P18 Cohort Study**

- 6-year cohort study examining factors influencing the health and well-being of 600 gay and bisexual men across the emerging adulthood period (i.e., ages 18–24).
- Participants were recruited from the New York metropolitan area.
- Eligibility criteria included: (a) 18-19 years of age at baseline; (b) identify as cisgender male; (c) self-report having had sex with a male in the past 6 months; and (d) self-reported negative HIV status at baseline

- **Study Procedures**

- Participants completed an audio computer-assisted self-interview (ACASI) baseline survey in order to provide information pertaining to socio-demographics, substance use, mental health, and psychosocial factors.
- Thereafter, participants completed follow-up assessments every 6 months where they were asked about substance use behaviors over the preceding 30 days using the timeline follow-back (TLFB) method.

Study Hypotheses

- Based on previous work, we hypothesized that the largest trajectory group in the alcohol and marijuana models would show an overall increase in alcohol and marijuana use in early emerging adulthood, then a general decline in late emerging adulthood.
- In addition we hypothesized that there would be additional trajectory groups that would have lower membership and would show patterns of excessive usage and no usage.

Models including the covariate of Race/Ethnicity

- We hypothesized that non-White YSMM would have lower trajectories of alcohol and marijuana use over the emerging adulthood period as compared to White YSMM.

Four Step Analytic Plan

Step	Description
1. Descriptive Statistics/Data Exploration	• Descriptive and bivariate statistics examined for each study variable. • Examination of overdispersion of zero values for alcohol and marijuana across the 14 waves.
2. Unadjusted Model(s): Censored Normal and Zero-Inflated Specifications	• Evidence of overdispersion was present for both alcohol and marijuana use; thus, we took two steps to account for this. • First, we took the square root of alcohol and marijuana use to try to “normalize” the outcomes and then ran GBTMs as if the outcomes were normal. • Second, we ran GBTMs with a zero-inflated specification to better capture the structure of the data generating process (DGP).
3. Determine the Best Number of Group(s) and Trajectory Shape(s)	• In this step, we used a combination of theory, previous research, and conceptual frameworks as well as statistical diagnostics to determine the ‘best’ number of groups and trajectory shapes for each model. • AIC and BIC were utilized for model selection, as well as posterior probabilities.
4. Adjusted Model(s)	• Two approaches for examining characteristics of group membership – (1) export trajectory assignments and use multinomial regression and (2) add covariate(s) to the GBTM models itself.

Study Sample Characteristics (N=597)

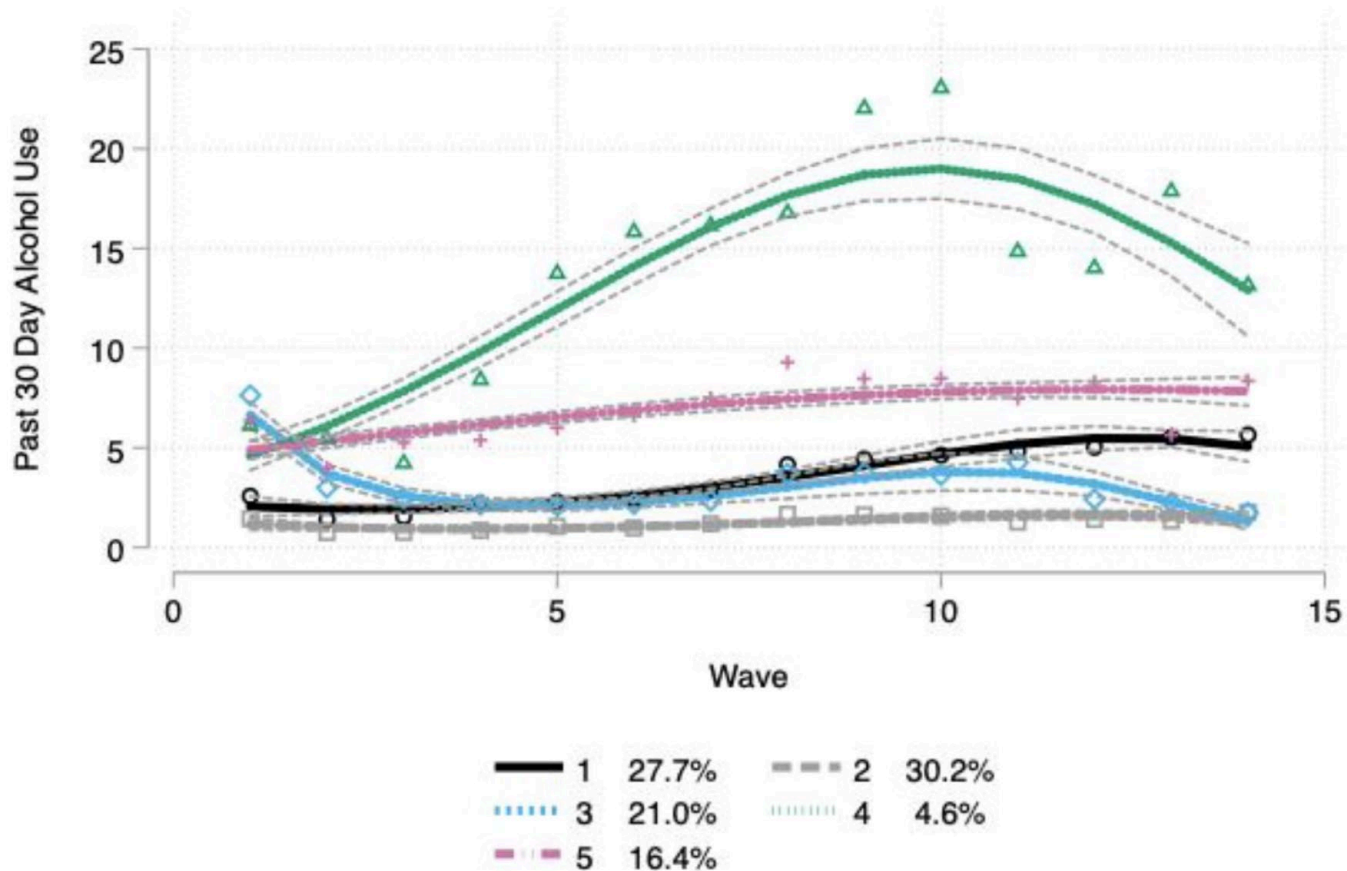
	%
Racial/ethnic identity	
White	29
Black	15
Hispanic	38
Other	18
Student status	
Enrolled in school at baseline	86
Not enrolled in school at baseline	14
Familial socioeconomic status	
Upper-class	4
Upper-middle class	25
Middle-class	37
Lower-middle class	24
Lower-class	10
Sexual orientation	
Exclusively homosexual	41
Predominately homosexual, incidentally heterosexual	29
Predominately homosexual, more than incidentally heterosexual	13
Equally heterosexual and homosexual	12
Predominately heterosexual, more than incidentally homosexual	3
Predominately heterosexual, incidentally homosexual	2

Results: Alcohol Use – Unadjusted Model



- We found the best fitting GBTM model to be the one with a zero-inflated specification
- The best-fitting model displayed 5 distinct trajectories, with 4 cubic polynomials and 1 quadratic polynomial:
 1. Low gradual increase drinker (28%)
 2. Consistently low drinkers (30%)
 3. Moderate drinkers who waiver over time and eventually decrease (21%)
 4. Moderate drinkers who become heavy drinkers (5%)
 5. Consistently moderate drinkers (16%)

Results: Alcohol Use – Unadjusted Model



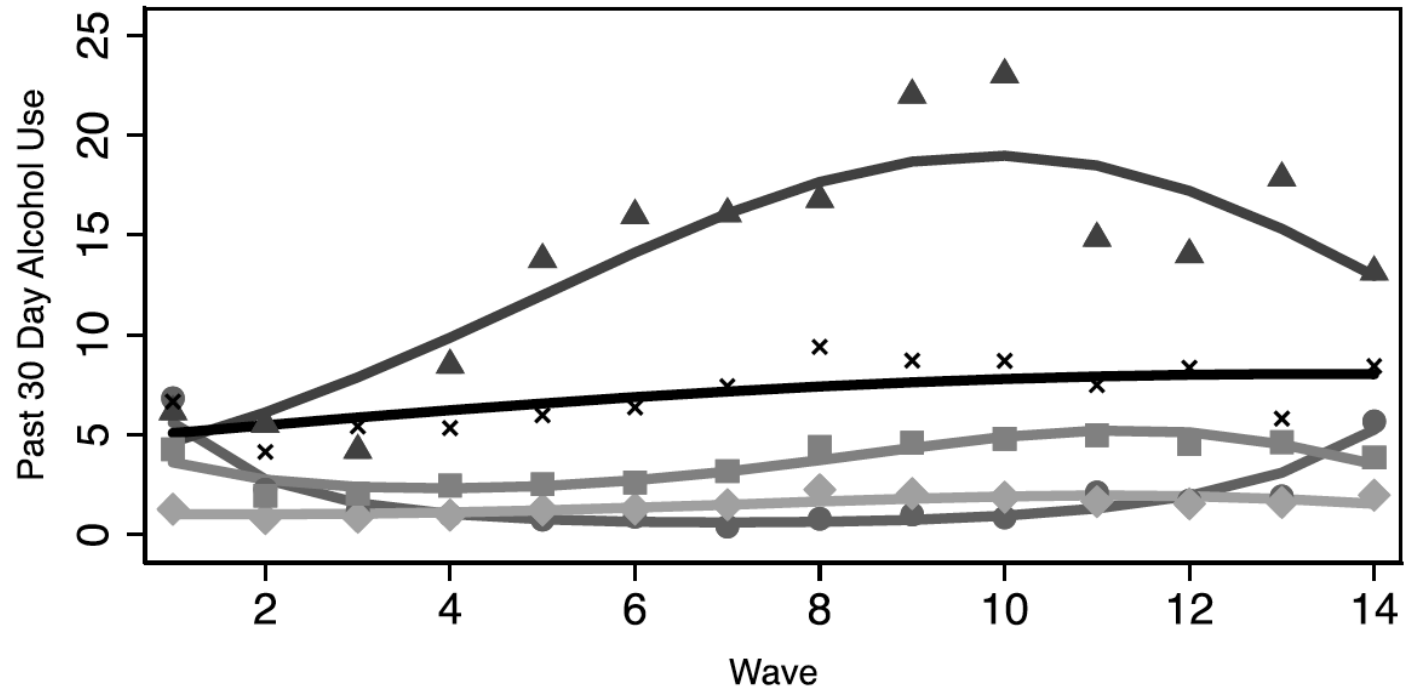
Note: Each “dot” represents the average number of days that an average individual in the trajectory group is expected to use alcohol over the preceding 30 days at each wave. Each line represents the average trajectory of alcohol use for each trajectory group. The gray lines around each line represent the confidence interval for each trajectory.

Results: Alcohol Use – Adjusted Model



- Prior to the addition of race/ethnicity as a covariate, we exported the predicted group membership profiles and ran a chi-square test of independence to examine if membership was associated with racial/ethnic identity.
- We then added racial/ethnic identity as a “risk factor” to the GBTM and found evidence for 5 distinct trajectories of alcohol use over time:
 1. Those who started out as moderate drinkers and who decreased over time (11%)
 2. Consistently low drinkers (32%)
 3. Those who start out as moderate drinkers who increase over time (36%)
 4. Those who start out as moderate drinkers who sharply increase over time (5%)
 5. Consistently moderate drinkers (17%)

Results: Alcohol Use – Adjusted Model



● 1	11.2%	◆ 2	32.0%
■ 3	35.7%	▲ 4	4.6%
× 5	16.5%		

Results: Alcohol Use – Adjusted Model



- Compared to White YSMM, Black (OR=6.5) and Hispanic (OR=5.7) YSMM and YSMM of another racial/ethnic identity (OR=58.6) had higher relative odds of being classified in the “consistently low” group as compared to the “moderate decrease” group.
- Further, YSMM classified in the “other” racial/ethnic identity category had higher relative odds of being classified in the “heavy increase” group as compared to the “moderate decrease” group (OR=16.0).
- YSMM classified in the “other” racial/ethnic identity category had higher relative odds of being classified in the “consistently moderate” group as compared to the “moderate decrease” group (OR=11.6).

Results: Marijuana Use – Unadjusted Model

- We found the best fitting GBTM model to be the one with a zero-inflated specification.
- The best-fitting model displayed 5 unique trajectories using a mixture of polynomials (1 linear term and 4 quadratic terms):
 1. Abstainers (45%)
 2. Moderate users who gradually increase in use (12%)
 3. Low users who drastically increased over time (11%)
 4. Moderate users who decreased over time (13%)
 5. Frequent users who increased to heavy use (20%)

Results: Marijuana Use – Unadjusted Model

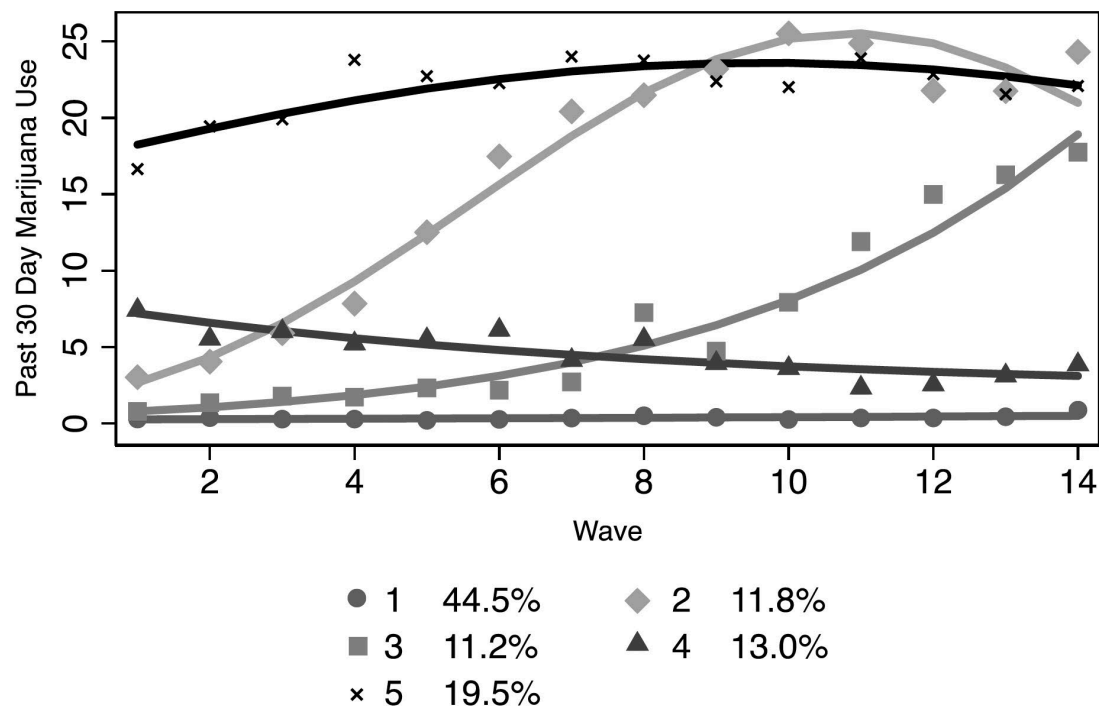


FIGURE 1

ZIP marijuana trajectories, unadjusted.

Note. Each “dot” represents the average number of days that an average individual in the trajectory group is expected to use marijuana over the preceding 30 days at each wave. Each line represents the average trajectory of marijuana use for each trajectory group.

Results: Marijuana Use – Adjusted Model



- Prior to adding race/ethnicity as a covariate, we exported the predicted group membership profiles and ran a chi-square test of independence to examine if membership was associated with racial/ethnic identity.
- We did not find enough evidence to suggest race/ethnicity was associated with group membership, thus we did not present this model.

Conclusion

- In particular, we found that transforming our outcomes with a square root transformation was insufficient to deal with the zero-inflation of alcohol and marijuana use over time.
- The ZIP specification allowed us to adjust for the extreme overdispersion of zero values and to identify trajectories that we may not have been able to capture in a censored normal model.
- Group-based trajectory analyses are most useful when guided by a substantive theoretical framework that can aid in the specifications of the size and number of trajectory shapes.

Conclusion, cont'd

- Further, we found that race/ethnicity predicted alcohol use trajectory membership among our sample of YSMM, such that those who identified as a racial/ethnic minority were more likely to be in groups of lower alcohol usage as compared to White YSMM
- However, we did not find evidence to suggest that race/ethnicity was associated with marijuana use trajectory membership among our study sample.